Sharp reset-error asymptotics for time-homogeneous qubit dynamics

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Autonomous qubit reset must remove unwanted population from one basis input while limiting the disturbance of the target input. For reset toward $|1\rangle$, let $a = P(0|0)$ and $\epsilon = P(0|1)$. The largest population-order reversal at fixed actual disturbance is $G(\epsilon) = \max(\epsilon - a)$, optimized over all time-independent qubit Lindblad generators and their probability closure. Its sharp asymptotic behavior is $G(\epsilon) \sim (4/e)\epsilon/\log(1/\epsilon)$ as $\epsilon \downarrow 0$. For generators admitting a stationary state diagonal in the preparation–measurement basis, the sharp law is $G_{\mathrm{diag}}(\epsilon) \sim (64/e^2)\epsilon/\log^2(1/\epsilon)$. Explicit single-jump constructions asymptotically attain both laws. The two scales arise from direct coherence readout in a tilted stationary eigenbasis and coherence feedback into population, respectively. Along every sequence of finite generators saturating the unrestricted law, every stationary state has squared coherence asymptotic to ϵ in the preparation–measurement basis and purity defect $o(\epsilon)$. These results quantify the role of stationary coherence in autonomous reset and determine sharp endpoint constraints on homogeneous quantum embedding.

I. INTRODUCTION

Autonomous reset prepares a target state using fixed open-system dynamics. For a qubit, one generator must remove unwanted population from one basis input while limiting the disturbance of the target input. Determining which pairs of conditional probabilities can arise from a time-independent quantum Markov evolution is a homogeneous quantum-embedding problem.

Shahbeigi *et al.* studied homogeneous quantum embedding of stochastic matrices and obtained analytic exclusion regions and numerical boundary candidates [1]. Their four-parameter single-jump family provides the antecedent for the attaining dynamics below, but its optimality over all generators remained unresolved. The results presented here determine the sharp reset-endpoint boundary of the full class, the distinct boundary for generators admitting a diagonal stationary state, and the stationary-state structure required for leading-order saturation.

The coherence mechanisms underlying these results have established precedents. Quantum fixed-point reachability can exceed classical population reachability [2], and classical reductions of quantum dynamics can lose P-divisibility as coherence returns information to populations [3]. Near reset, the results below quantify the largest population-order reversal permitted by a single time-independent generator. Explicit constructions identify the mechanisms behind the two asymptotic scales, while uniform upper bounds establish their optimality over the corresponding generator classes.

II. MODEL

Prepare a qubit in either $|0\rangle$ or $|1\rangle$, evolve both inputs with the same generator for the same duration, and measure in the preparation basis. Near reset toward $|1\rangle$, write the column-stochastic probability table as

$$T(a, 1-\epsilon) = \begin{pmatrix} a & \epsilon \\ 1-a & 1-\epsilon \end{pmatrix}. \tag{1}$$

Here a is the unwanted output population for input $|0\rangle$, and ϵ is the disturbance of the target input $|1\rangle$. Their difference equals the determinant of the probability table:

$$q := a - \epsilon = \det T. \tag{2}$$

For a diagonal input with unwanted population r, linearity gives

$$\rho_r = r|0\rangle\langle 0| + (1-r)|1\rangle\langle 1|, \qquad P_0(r) = \epsilon + (a-\epsilon)r. \tag{3}$$

Thus $q < 0$ means that an input with more unwanted population ends with less of it. The optimization problem is to determine the largest such population-order reversal generated by one fixed qubit Lindbladian.

Fixing the actual measured disturbance ϵ selects a slice of the homogeneous-embedding region. For a realized reversal $\Delta = \epsilon - a > 0$, the worst population error among diagonal inputs remains ϵ, while the error averaged equally over the two basis inputs is $\epsilon - \Delta/2$. This is therefore an optimization of input-conditioned population data. With only an upper allowance $\epsilon \le \delta$ and unrestricted time and rates, ordinary damping toward $|1\rangle$ could already approach perfect reset with $\epsilon = 0$. The two measured probabilities do not specify the response to coherent inputs.

The model includes all time-independent Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) generators [4, 5]

$$\begin{aligned} \mathcal{L}(\rho) &= -i[H,\rho] + \sum_{\mu} \mathcal{D}[J_\mu](\rho), \\ \mathcal{D}[J](\rho) &= J\rho J^\dagger - \tfrac{1}{2}\{J^\dagger J, \rho\}, \end{aligned} \tag{4}$$

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on $\mathbb{C}^2$, with a Hermitian Hamiltonian and a finite collection of finite jump matrices. No auxiliary levels are allowed, and the preparation–measurement basis remains fixed. Rates, duration, and Hamiltonian strength are unrestricted.

Identify each table $T(a, 1-\epsilon)$ with the pair $(a, 1-\epsilon)$. Let $\mathcal{Q}_2^{\mathrm{fin}}$ be the tables obtained from $e^{t\mathcal{L}}$ at finite $t \geq 0$, and set

$$\mathcal{Q}_2 = \overline{\mathcal{Q}_2^{\mathrm{fin}}} \subset [0,1]^2. \tag{5}$$

This probability closure implements the arbitrary-accuracy convention of Ref. [1]: each approximant has finite parameters, with no common bound on rates or duration. Rescaling $H \mapsto tH$, $J_\mu \mapsto \sqrt{t} J_\mu$ fixes the observation time at one and retains singular parameter sequences.

At fixed actual disturbance, define

$$G(\epsilon) = \max\{\epsilon - a : (a, 1-\epsilon) \in \mathcal{Q}_2\}. \tag{6}$$

The maximum exists because $\mathcal{Q}_2$ is compact and its fixed-ϵ slice is nonempty: classical relaxation to population ϵ gives $a = \epsilon$ in the closure. In particular, $0 \leq G(\epsilon) \leq \epsilon$.

Let $\mathcal{Q}_{\mathrm{diag}}$ be the probability closure of finite realizations whose generator admits at least one stationary state diagonal in the preparation–measurement basis. This class permits transient coherence and population–coherence coupling; a unique stationary state and preservation of all diagonal states are not required. Define G_{diag} by (6) with $\mathcal{Q}_2$ replaced by $\mathcal{Q}_{\mathrm{diag}}$.

III. RESET-ERROR BOUNDS

The two classes have different asymptotic boundaries near perfect reset.

Theorem 1 (Sharp autonomous reset law). For the full time-independent qubit GKSL probability closure,

$$\lim_{\epsilon \downarrow 0} \frac{\log(1/\epsilon)}{\epsilon} G(\epsilon) = \frac{4}{e}. \tag{7}$$

Equivalently, the lower reset boundary is

$$\min\{a : (a, 1-\epsilon) \in \mathcal{Q}_2\} = \epsilon \left[1 - \frac{4/e + o(1)}{\log(1/\epsilon)}\right]. \tag{8}$$

Theorem 2 (Diagonal-stationary-state law). The restricted class has the sharp endpoint asymptotic

$$G_{\mathrm{diag}}(\epsilon) = \left(\frac{64}{e^2} + o(1)\right) \frac{\epsilon}{\log^2(1/\epsilon)}. \tag{9}$$

Both laws imply $a/\epsilon \to 1$ on the optimal boundary. The two input-conditioned errors agree to leading order, while their smaller difference $\epsilon - a$ measures the population-order reversal. This difference is a vanishing fraction of the actual disturbance in the perfect-reset limit.

The two asymptotic laws lie between the classical and time-dependent quantum limits:

$$\begin{aligned} &G_{\mathrm{cl}}(\epsilon) = 0, && G_{\mathrm{diag}}(\epsilon) \sim \frac{64}{e^2} \frac{\epsilon}{\log^2(1/\epsilon)}, \\ &G(\epsilon) \sim \frac{4}{e} \frac{\epsilon}{\log(1/\epsilon)}, && G_{\mathrm{td}}(\epsilon) = \epsilon. \end{aligned} \tag{10}$$

Here G_{cl} and G_{td} denote the same fixed-error optimization, with the same probability-closure convention, for classical two-state dynamics and time-dependent qubit GKSL control, respectively. Even with time-dependent classical rates, $q(t) = \exp[-\int_0^t (k_{01}(s) + k_{10}(s))\, ds] \geq 0$. Time-dependent qubit GKSL control permits every two-state stochastic table [2]. All four comparisons use the same system dimension, preparation–measurement basis, and probability task.

The unrestricted leading optimum also constrains the stationary states of any attaining sequence. For a finite realization with $a \leq \epsilon$ sufficiently small, write any stationary state in its eigenbasis as

$$\begin{aligned} &\rho_* = \nu |e\rangle\langle e| + (1-\nu)|g\rangle\langle g|, \\ &0 \leq \nu < \tfrac{1}{2}, \qquad p = |\langle 0|g\rangle|^2. \end{aligned} \tag{11}$$

Here ν is the smaller stationary eigenvalue, and p measures the tilt of the dominant eigenvector away from the target state $|1\rangle$.

Corollary 3 (Stationary-state saturation). Consider a sequence of finite qubit GKSL realizations with actual errors a_n, ϵ_n, where $\epsilon_n \downarrow 0$ and

$$\frac{\log(1/\epsilon_n)}{\epsilon_n}(\epsilon_n - a_n) \longrightarrow \frac{4}{e}. \tag{12}$$

For each sufficiently large n, choose any stationary state and let ν_n, p_n be as in (11). Then

$$\frac{\nu_n}{\epsilon_n} \longrightarrow 0, \qquad \frac{p_n}{\epsilon_n} \longrightarrow 1. \tag{13}$$

In the fixed measurement basis this gives

$$\begin{aligned} |\langle 0|\rho_{*,n}|1\rangle|^2 &= (1-2\nu_n)^2 p_n (1-p_n) \sim \epsilon_n, \\ 1 - \mathrm{tr}(\rho_{*,n}^2) &= 2\nu_n(1-\nu_n) = o(\epsilon_n). \end{aligned} \tag{14}$$

Although stationary coherence vanishes in the reset limit, its square exhausts the leading population-error budget. The purity defect is of smaller order. Section V proves these necessary conditions for sequences of finite realizations.

The relative separation of the two optimal gains is

$$\frac{G_{\mathrm{diag}}(\epsilon)}{G(\epsilon)} \sim \frac{16}{e \log(1/\epsilon)} \longrightarrow 0. \tag{15}$$

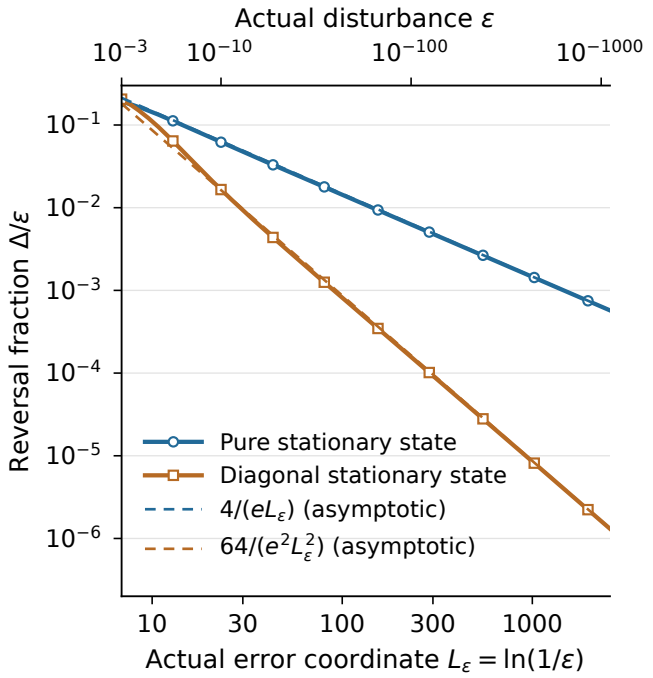


FIG. 1. Population-order reversal at fixed actual disturbance $\epsilon = P(0|1)$, shown as Δ/ϵ with $\Delta = \epsilon - a$. The horizontal coordinate is $L_\epsilon = \log(1/\epsilon)$; the upper axis gives ϵ. Solid curves with circles and squares show the tilted pure-stationary-state construction, Eqs. (17)–(19), and the diagonal-stationary-state construction, Eqs. (C15)–(C18), respectively. Dashed curves show the leading laws $4/(eL_\epsilon)$ and $64/(e^2L_\epsilon^2)$. Both axes are logarithmic. The solid curves are attainable lower bounds for the corresponding optimal gains; markers distinguish the constructions. At $\epsilon = 10^{-3}$ the diagonal construction exceeds the displayed pure-state construction. The hierarchy of Theorems 1 and 2 concerns the optimized gains as $\epsilon \to 0$.

Consequently any finite-generator sequence for which $\liminf \log(1/\epsilon)(\epsilon - a)/\epsilon > 0$ eventually admits no stationary state diagonal in the measurement basis: a contrary subsequence would satisfy Theorem 2 and have vanishing scaled gain. Stationary coherence is therefore necessary for a nonvanishing fraction of the unrestricted leading scale, while transient coherence can realize the smaller scale.

IV. ASYMPTOTICALLY OPTIMAL CONSTRUCTIONS

The two bounds are attained through different population–coherence mechanisms. Figure 1 compares explicit realizations at the same actual disturbance ϵ. These curves give constructive lower bounds; their ordering at finite error can differ from that of the optimized classes.

A. Tilted pure stationary state

For a stationary eigenvector with overlap $p = |\langle 0|g\rangle|^2$, define the readout coefficients

$$\begin{aligned} v &= 1 - 2p, \qquad w = 2\sqrt{p(1-p)}, \\ V &= v^2, \quad W = w^2, \quad R = vw. \end{aligned} \tag{16}$$

Let $L \to \infty$ and define

$$p = e^{-L}, \qquad A = L + \log L + 1 - \log 4, \qquad B = A/L. \tag{17}$$

Choose orthonormal vectors

$$|g\rangle = \sqrt{p}|0\rangle + \sqrt{1-p}|1\rangle, \qquad |e\rangle = \sqrt{1-p}|0\rangle - \sqrt{p}|1\rangle, \tag{18}$$

and, in the ordered basis (e, g), take a single jump with no Hamiltonian:

$$J = \begin{pmatrix} -\sqrt{B} & 0 \\ \sqrt{A} & 0 \end{pmatrix}, \qquad H = 0. \tag{19}$$

For every sufficiently large finite L, this generator has stationary state $|g\rangle\langle g|$. For a state ρ, write $n = \langle e|\rho|e\rangle$ for its population and $z = \langle e|\rho|g\rangle$ for its coherence in the stationary eigenbasis. The measured population is $P_0 = p + vn + w\,\mathrm{Re}\,z$, and the dynamics obeys

$$\dot{n} = -An, \qquad \dot{z} = -\sqrt{AB}\,n - \Gamma z, \qquad \Gamma = (A+B)/2. \tag{20}$$

Set

$$\begin{aligned} F &= e^{-A}, \qquad S = e^{-\Gamma}, \\ J_0 &= \int_0^1 e^{-At} e^{-\Gamma(1-t)}\,dt = \frac{S-F}{A-\Gamma}. \end{aligned} \tag{21}$$

For the difference between the two input states, let u and z denote the population and coherence differences in these coordinates. Then $q = vu + w\,\mathrm{Re}\,z$, with $(u(0), z(0)) = (v, w)$, and direct solution gives

$$q = v^2F + w^2S - vw\sqrt{AB}\,J_0. \tag{22}$$

For the target input $|1\rangle$, the initial conditions are $n(0) = p$, $z(0) = -w/2$, giving the actual disturbance

$$\epsilon(L) = p + vpF - \frac{w^2}{2}S - pw\sqrt{AB}\,J_0. \tag{23}$$

In (22), the surviving population and coherence terms are positive, while population transfer into coherence produces the negative cross term. The tilt of the stationary eigenbasis makes this coherence directly visible in the population measurement. With $A \sim L$, $B/A = 1/L$, and $\epsilon \sim e^{-L}$, the negative cross term has twice the magnitude of the surviving population term to leading order, giving $\Delta \sim (4/e)\epsilon/\log(1/\epsilon)$. Appendix B 5 derives these asymptotics and shows that the family realizes every sufficiently small positive actual disturbance ϵ.

The representation (19) has $H = 0$, while the jump contains coherent amplitudes. Identity shifts of the jump redistribute terms between the dissipator and Hamiltonian, so the value of H alone does not characterize the coherent resources of the dynamics.

The same construction permits fixed jump strength. Since the squared operator norm is $\|J\|^2 = A + B$, any fixed rate $\kappa > 0$ gives the rescaling

$$\begin{aligned} J_{\mathrm{phys}} &= \sqrt{\frac{\kappa}{A+B}}J, \\ t_{\mathrm{phys}} &= \frac{A+B}{\kappa} \\ &= \frac{\log(1/\epsilon) + \log\log(1/\epsilon) + O(1)}{\kappa}. \end{aligned} \tag{24}$$

The rescaled dynamics gives the same channel with $\|J_{\mathrm{phys}}\|^2 = \kappa$. The stationary-state direction and ratio B/A depend on the target error, while each chosen generator remains fixed during its evolution.

B. Diagonal stationary state

When $\rho_* = \mathrm{diag}(\nu, 1-\nu)$ in the measurement basis, stationarity gives

$$a = \nu + (1-\nu)q, \qquad \epsilon = \nu(1-q). \tag{25}$$

For $q = -\Delta < 0$, these relations become

$$\nu - a = (1-\nu)\Delta, \qquad \epsilon - \nu = \nu\Delta. \tag{26}$$

The unwanted population for input $|0\rangle$ transiently falls below its stationary value, while the target input ends slightly above that value. Classical two-state relaxation cannot cross the stationary population.

For the difference $\delta\rho$ between the two input trajectories, let $u = \delta\rho_{00}$ and $z = \delta\rho_{01}$ in the measurement basis. A single jump and a Hamiltonian can keep z real and realize

$$\frac{d}{dt}\begin{pmatrix} u \\ z \end{pmatrix} = \begin{pmatrix} -\alpha & r_0 \\ -c_0 & -\lambda_+ \end{pmatrix}\begin{pmatrix} u \\ z \end{pmatrix}, \qquad \lambda_+ = \frac{A(1+\chi)^2 + B}{2}, \tag{27}$$

where $\chi = \sqrt{\nu/(1-\nu)}$ and $\alpha, r_0, c_0, \lambda_+ > 0$. Appendices A and C specify the coefficients and the explicit GKSL realization. The input difference starts at $(u, z) = (1, 0)$, and the measured determinant is $q = u(1)$. Population first generates coherence through $-c_0 u$, which then feeds back into population through $r_0 z$. This negative feedback can transiently reverse the population difference even when stationary coherence is zero.

For both asymptotically optimal constructions, $\tau = B/A$ is of order $1/\log(1/\epsilon)$. The negative slow-mode amplitude is then $8\sqrt{\nu}\,\tau$ to leading order for the diagonal construction, compared with $4\sqrt{p\tau}$ for direct readout in the tilted pure-state construction. The competing positive population mode decays about twice as fast. Optimizing their difference gives, to leading order, one quarter of the squared slow-mode amplitude multiplied by $e^{-\tau L}$. The exponential factor comes from the $O(\tau)$ correction to the slow-to-fast decay-rate ratio $1/2$, with $L = \log(1/\nu)$ or $\log(1/p)$, respectively. The resulting leading gains are $16\nu\tau^2 e^{-\tau L}$ and $4p\tau e^{-\tau L}$, maximized at $\tau L = 2$ and $\tau L = 1$. The extra factor of τ accounts for the additional inverse logarithm in the diagonal case. Appendix C proves the uniform upper bound for this class and gives a matching GKSL realization. In that family, $\nu - a \sim (64/e^2)\nu/\log^2(1/\nu)$.

Thus a diagonal stationary state permits transient coherence generated by $-c_0 u$. The two theorems quantify the distinct gains available from this feedback and from direct readout enabled by stationary coherence.

V. UNIVERSAL UPPER BOUNDS

Optimality follows from an upper bound valid for arbitrary generators. A stationary-state constraint bounds the available error budget, and a population–coherence comparison bounds the transient reversal.

A. Stationary-state constraint

Every finite-dimensional GKSL generator has a stationary density matrix, obtained from a convergent subsequence of long-time orbit averages. For any fixed point ρ_* of the channel $\Phi = e^{\mathcal{L}}$, set $p_* = \langle 0|\rho_*|0\rangle$. The positive effect $\Phi^*(|0\rangle\langle 0|)$ has diagonal entries a, ϵ, so the magnitude of its off-diagonal entry is at most $\sqrt{a\epsilon}$. Positivity of ρ_* and stationarity give $p_* \le (\sqrt{ap_*} + \sqrt{\epsilon(1-p_*)})^2$, and therefore

$$[(1-\sqrt{a})^2 + \epsilon]p_* \le \epsilon. \tag{28}$$

With (11), $p_* = \nu + (1-2\nu)p$. In the relevant region $a \le \epsilon \downarrow 0$, the exact inversion in Appendix A yields

$$p + \nu \le \varrho(\epsilon) = \epsilon[1 + O(\sqrt{\epsilon})]. \tag{29}$$

Stationary-state tilt and mixedness share the same small measured-error budget. Both must be retained to control the finer scale $\epsilon/\log(1/\epsilon)$.

B. Population–coherence comparison

In an arbitrary stationary eigenbasis, set $u = \langle e|\rho|e\rangle - \nu$ and $z = \langle e|\rho|g\rangle$. Their homogeneous equations have the form

$$\dot{u} = -\alpha u + \mathrm{Re}(R_0 z^*), \qquad \dot{z} = \widetilde{C}u + Ez^* - (\Gamma + i\omega)z. \tag{30}$$

Appendix A derives these equations for arbitrary jump collections and Hamiltonians and gives the exact coefficients in (A7)–(A8). The equations retain both population feedback and coupling to the conjugate coherence.

The frequency term $-i\omega z$ does not contribute directly to the derivative of $|z|$, making the comparison uniform in ω. Gram positivity bounds the coupling strengths by nonnegative constants r_0, c_0 and the coherence damping from below by λ. These bounds define the comparison matrix $K = \left(\begin{smallmatrix} -\alpha & r_0 \\ c_0 & -\lambda \end{smallmatrix}\right)$ with nonnegative off-diagonal entries and parameters given in (A11). Choose the relative phase of e, g so that the off-diagonal entry of $|0\rangle\langle 0|$ is nonnegative. The input difference obeys the same equations, with $(u(0), z(0)) = (v, w)$ and $q = vu + w\,\mathrm{Re}\,z$, using (16). Retaining the positive free decay of the population gives

$$q(t) \geq 2Ve^{-\alpha t} - (v, w)e^{tK}\begin{pmatrix} v \\ w \end{pmatrix} \qquad (t \geq 0). \tag{31}$$

The proof in Appendix A first bounds $|u|, |z|$ by a positive comparison system, then applies the variation-of-constants formula to the population. This step preserves the positive term competing with the negative coherence transient.

The rate parameters A, B are defined in (A4) from the jump entries in a stationary eigenbasis, after an identity shift sets the gg entries to zero. For the pure-state construction, they coincide with the rates in (19). At time one, define

$$\mathcal{E}(p, \nu, A, B) = \left[(v, w)e^{K}\begin{pmatrix} v \\ w \end{pmatrix} - 2Ve^{-\alpha}\right]_+, \qquad [x]_+ = \max(x, 0). \tag{32}$$

Then $(\epsilon - a)_+ \leq \mathcal{E}$. The positive comparison system enlarges the physical class by bounding its couplings independently.

Proposition 4 (Sharp uniform envelope). Let $p, \nu \geq 0$, $p + \nu \leq \rho$, and $A, B \geq 0$. With the definitions above, as $\rho \downarrow 0$,

$$\mathcal{E}(p, \nu, A, B) \leq \frac{4p/e + o(\rho)}{\log(1/\rho)}. \tag{33}$$

The remainder is a function of ρ alone and is uniform in all four parameters.

The leading contribution comes from a small window of the rate ratio $\tau = B/A$. With $L = \log(1/\rho)$, optimizing the overall rate and using the budget constraint $p + \nu \leq \rho$ give

$$\mathcal{E} \leq [4p + o(\rho)]\tau e^{-\tau L} \leq \frac{4p/e + o(\rho)}{L}. \tag{34}$$

The factor $\tau e^{-\tau L}$ is maximized at $\tau L = 1$. Appendix B establishes this estimate uniformly and bounds the remaining rate ratios, including zero rates, repeated eigenvalues, and arbitrarily large rates. Their contributions are $o(\rho/L)$.

C. Sharp asymptotics and saturation

For a finite realization with $a \leq \epsilon$, choose any stationary state. Lemma 5 gives $p + \nu \leq \varrho(\epsilon)$, and Lemma 6 and Proposition 4 then yield

$$\epsilon - a \leq \frac{4p/e + o(\varrho(\epsilon))}{\log(1/\varrho(\epsilon))} \leq \left(\frac{4}{e} + o(1)\right)\frac{\epsilon}{\log(1/\epsilon)}. \tag{35}$$

The last step uses $\varrho(\epsilon) = \epsilon[1 + O(\sqrt{\epsilon})]$. If $a > \epsilon$, the upper bound is immediate.

Uniformity means that, for every $\delta > 0$, there is an $\epsilon_\delta > 0$ such that every finite generator obeys

$$\epsilon - a \leq \left(\frac{4}{e} + \delta\right)\frac{\epsilon}{\log(1/\epsilon)} \qquad (0 < \epsilon < \epsilon_\delta). \tag{36}$$

Fix a point in the probability closure with $0 < \epsilon < \epsilon_\delta$ and approximate it by finite realizations. The approximants eventually lie in the same error interval. Continuity of the right-hand side in ϵ then extends (36) to $\mathcal{Q}_2$.

The pure-state construction in Section IV supplies the matching lower bound, with its actual disturbance and coverage established in Appendix B 5, completing Theorem 1. The refined coefficient $4p/e$ also proves Corollary 3:

Proof. For all sufficiently large n, $a_n < \epsilon_n$. The budget gives $p_n + \nu_n \leq \epsilon_n[1 + o(1)]$. The refined first inequality of (35), multiplied by $\log(1/\epsilon_n)/\epsilon_n$, and (12) force $\liminf p_n/\epsilon_n \geq 1$. Both limits follow from nonnegativity and the budget. □

The coherence and purity identities in (14) follow directly from the stationary eigenvalues and the overlap p_n.

For the diagonal class, $p = 0$ leaves only feedback from population through coherence and back into population. Appendix C refines the comparison to the scale $\nu/\log^2(1/\nu)$, proves a uniform bound over all rate ratios, and constructs a matching GKSL generator. Equation (25) converts ν to the actual disturbance ϵ, and the same probability-closure argument completes Theorem 2.

VI. DISCUSSION

The two reset laws constrain the conditional population probabilities compatible with one fixed generator. Their logarithmic restriction reflects autonomous evolution. Time-dependent control can first damp toward $|0\rangle$, leaving population ϵ in $|1\rangle$ for input $|1\rangle$, and then apply a unitary swap. This gives $a = 0$ and $P(0|1) = \epsilon$. With the same system dimension and preparation–measurement basis, time-dependent control therefore reaches $G_{\mathrm{td}}(\epsilon) = \epsilon$, while a fixed generator reaches a vanishing fraction of this value.

Within autonomous dynamics, the two scales distinguish direct coherence readout from coherence feedback

into population. The diagonal-stationary-state law quantifies transient overshoot beyond classical relaxation [2, 3]. Transient coherence suffices to produce this overshoot. Saturating the unrestricted law requires squared stationary coherence to exhaust the leading population-error budget and the purity defect to be of smaller order. These conditions hold for every stationary state along an asymptotically optimal sequence of finite GKSL realizations.

Both attaining families are contained in the earlier four-parameter ansatz or its probability closure. That ansatz uses $H = X$, where X is the Pauli x matrix, and a rank-one jump between states of the form $(\cos\theta, i\sin\theta)^T$ [1]. The pure-state family lies in its probability closure, while the diagonal-state family belongs to the ansatz itself at sufficiently small error. Appendix D gives the transformations. The upper bounds for the two generator classes establish the endpoint optimality of these families.

The earlier exclusion bound implies $a \geq (\epsilon/2)^{10}[1+o(1)]$ near reset. Theorem 1 determines $a/\epsilon \to 1$ and its leading correction. Appendix E exhibits the additional excluded region $\epsilon/2 < a < 3\epsilon/4$ for sufficiently small ϵ.

The finite-error optimal curve, uniform higher-order corrections, and the boundary away from reset remain open.

VII. CONCLUSIONS

For autonomous qubit reset, the largest population-order reversal at fixed actual disturbance ϵ obeys $G(\epsilon) \sim (4/e)\epsilon/\log(1/\epsilon)$. Requiring a stationary state diagonal in the preparation–measurement basis reduces the sharp scale to $G_{\mathrm{diag}}(\epsilon) \sim (64/e^2)\epsilon/\log^2(1/\epsilon)$. Uniform upper bounds and explicit single-jump constructions establish both laws. Saturation of the unrestricted law further determines the stationary coherence and purity scales, connecting the observable reset-error boundary to the stationary-state structure required to attain it.

ACKNOWLEDGMENTS

[ACKNOWLEDGMENTS AND FUNDING TO BE COMPLETED]

DATA AND CODE AVAILABILITY

[DATA AND CODE AVAILABILITY TO BE CONFIRMED]

DECLARATIONS

[DECLARATIONS TO BE CONFIRMED]

Appendix A: Stationary-state reduction

This appendix derives the stationary error budget and the population–coherence comparison used in Section V.

1. Stationary-state constraint

Every finite-dimensional GKSL generator has a stationary density matrix. Compactness gives a convergent subsequence of the time averages of any orbit, and $\mathcal{L}(t^{-1}\int_0^t e^{s\mathcal{L}}(\rho)\,ds) = (e^{t\mathcal{L}}(\rho)-\rho)/t \to 0$. The following lemma bounds every such state in terms of the two measured errors.

Lemma 5 (Stationary error budget). Let Φ be a qubit channel with the probabilities in (1). For any fixed point ρ_* of Φ, write $p_* = \langle 0|\rho_*|0\rangle$. Then the inequality (28) holds. If $a \leq \epsilon < 1/4$, diagonalize ρ_* as in (11). Then $p, \nu < 1/2$ and

$$p_* = \nu + (1-2\nu)p, \tag{A1}$$

$$p + \nu \leq \varrho(\epsilon) := 1 - \sqrt{1-2b_*(\epsilon)},$$

$$b_*(\epsilon) = \frac{\epsilon}{1-2\sqrt{\epsilon}+2\epsilon}, \tag{A2}$$

$$\varrho(\epsilon) = \epsilon[1+O(\sqrt{\epsilon})].$$

Proof. The positive effect $F = \Phi^*(|0\rangle\langle 0|)$ has diagonal entries a, ϵ, so $|F_{01}| \leq \sqrt{a\epsilon}$. Positivity of ρ_* similarly bounds its off-diagonal entry. Since $\Phi(\rho_*) = \rho_*$,

$$p_* = \mathrm{tr}(F\rho_*) \leq \left(\sqrt{ap_*} + \sqrt{\epsilon(1-p_*)}\right)^2.$$

Taking square roots and rearranging proves (28). For $a \leq \epsilon < 1/4$, it gives $p_* \leq b_*(\epsilon) < 1/2$. Equation (A1) implies $\nu \leq p_*$ and $p \leq p_*$, so $s = p + \nu < 1$. Since

$$p_* = s - 2\nu p \geq s - \frac{s^2}{2},$$

and the right-hand side increases on $[0, 1)$, inversion yields (A2). □

2. Population–coherence dynamics

Fix an arbitrary stationary state and use the eigenbasis (e, g) from (11). Subtracting each jump's gg entry times the identity, with the corresponding Hamiltonian adjustment, preserves the generator and gives

$$J_\mu = \begin{pmatrix} x_\mu & y_\mu \\ z_\mu & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} \omega/2 & h \\ h^* & -\omega/2 \end{pmatrix}. \tag{A3}$$

An irrelevant scalar multiple of the identity has also been removed from H. Define

$$A = \sum_\mu |z_\mu|^2, \quad B = \sum_\mu |x_\mu|^2, \quad U = \sum_\mu |y_\mu|^2,$$

$$C = \sum_\mu x_\mu z_\mu^*, \quad D = \sum_\mu x_\mu^* y_\mu, \quad E = \sum_\mu y_\mu z_\mu^*. \tag{A4}$$

These coefficients form the positive semidefinite Gram matrix

$$\begin{pmatrix} B & D^* & C \\ D & U & E \\ C^* & E^* & A \end{pmatrix} \succeq 0, \tag{A5}$$

whose principal minors imply $|C| \le \sqrt{AB}$, $|D| \le \sqrt{BU}$, and $|E| \le \sqrt{AU}$. Stationarity gives the exact relations

$$U = \frac{A\nu}{1-\nu}, \qquad h = -i\frac{\nu C - D/2}{1-2\nu}. \tag{A6}$$

The real frequency ω remains unrestricted.

Let $u = \langle e|\rho|e\rangle - \nu$ be the $|e\rangle$ population relative to stationarity and $z = \langle e|\rho|g\rangle$ the coherence. Direct substitution in (4) gives

$$\dot{u} = -(A+U)u + \mathrm{Re}(R_0 z^*), \tag{A7}$$

$$\dot{z} = \widetilde{C}u + Ez^* - (\Gamma + i\omega)z, \tag{A8}$$

$$\widetilde{C} = \frac{C-D}{1-2\nu},$$

$$R_0 = \frac{2[(1-\nu)D - \nu C]}{1-2\nu},$$

$$\Gamma = \frac{A+B+U}{2}.$$

The same homogeneous equations govern the difference between two trajectories, including the population-feedback term in (A7) and the conjugate-coherence term in (A8). For $\nu = 0$, stationarity forces $U = D = E = R_0 = h = 0$, reducing the dynamics to a triangular population–coherence system.

Choose the relative phase of e, g so that the off-diagonal entry of $|0\rangle\langle 0|$ is nonnegative. Set v, w, V, W, R as in (16). In particular, $V + W = 1$. The difference between the two preparation-basis inputs is

$$|0\rangle\langle 0| - |1\rangle\langle 1| = \begin{pmatrix} v & w \\ w & -v \end{pmatrix}, \qquad (u(0), z(0)) = (v, w), \tag{A9}$$

and the measured determinant at time t is

$$q(t) = vu(t) + w\,\mathrm{Re}\,z(t). \tag{A10}$$

The basis change therefore transforms the preparation and measurement axes together.

3. Positive comparison

Introduce

$$\chi = \sqrt{\frac{\nu}{1-\nu}}, \quad \alpha = A(1+\chi^2), \quad \lambda = \frac{A(1-\chi)^2 + B}{2},$$

$$c_0 = \frac{\sqrt{AB}(1+\chi^2)}{1-\chi}, \quad r_0 = \frac{2\chi\sqrt{AB}}{1-\chi},$$

$$K = \begin{pmatrix} -\alpha & r_0 \\ c_0 & -\lambda \end{pmatrix}. \tag{A11}$$

The Gram inequalities and (A6) imply

$$\Gamma - |E| \ge \lambda, \qquad |\widetilde{C}| \le c_0, \qquad |R_0| \le r_0. \tag{A12}$$

These bounds use necessary consequences of Gram positivity. Their extremal values need not be simultaneously attainable by a physical generator.

Lemma 6 (Positive comparison). For $0 \le p \le 1/2$ and $0 \le \nu < 1/2$, every evolution above satisfies (31). This includes zero rates and arbitrary ω.

Proof. Equations (A7)–(A12) give upper right Dini-derivative bounds

$$D^+|u| \le -\alpha|u| + r_0|z|, \qquad D^+|z| \le c_0|u| - \lambda|z|.$$

The term $-i\omega z$ contributes zero to the derivative of $|z|$. At a zero of u or z, the corresponding modulus derivative gives the same inequality. Since K has nonnegative off-diagonal entries, its exponential has nonnegative entries. The componentwise comparison principle therefore gives $|u| \le X$ and $|z| \le Z$ for $(X, Z)^T = e^{tK}(v, w)^T$. This principle follows by adding a strictly positive perturbation to the comparison initial data and derivative, excluding a first contact, and then removing the perturbation.

The variation-of-constants formula for the population gives the stronger lower bound

$$\begin{aligned} u(t) &\ge ve^{-\alpha t} - r_0 \int_0^t e^{-\alpha(t-s)} Z(s)\, ds \\ &= 2ve^{-\alpha t} - X(t). \end{aligned}$$

Substitute this and $\mathrm{Re}\, z \ge -Z$ into (A10); $v, w \ge 0$ yields (31). □

The free population decay in (31) supplies the positive term that competes with the coherence transient; replacing u by $-|u|$ would discard this term. At time one, the comparison gives the envelope (32), with $(\epsilon - a)_+ \le \mathcal{E}$.

Appendix B: Uniform envelope bound

The proof of Proposition 4 separates four ranges of the rate ratio. Throughout this appendix, C_{b} denotes a finite universal constant that may increase between estimates. All thresholds depend only on ρ.

1. Dimensionless formulation

If $A = 0$, then $r_0 = c_0 = \alpha = 0$ and $\mathcal{E} = [We^{-B/2} - V]_+ = 0$ for small ρ. For $A > 0$, write

$$\tau = B/A, \qquad \mathfrak{a} = 1 + \chi^2, \qquad \ell = \frac{(1-\chi)^2 + \tau}{2},$$

$$c = \frac{\sqrt{\tau}(1+\chi^2)}{1-\chi}, \quad r = \frac{2\chi\sqrt{\tau}}{1-\chi}, \quad h_0 = rc, \tag{B1}$$

so that $K = AM$ with $M = \left(\begin{smallmatrix} -\mathfrak{a} & r \\ c & -\ell \end{smallmatrix}\right)$. Its decay rates are

$$\beta = \frac{\mathfrak{a}+\ell-d}{2}, \qquad \gamma = \frac{\mathfrak{a}+\ell+d}{2}, \qquad d = \sqrt{(\mathfrak{a}-\ell)^2+4h_0}. \tag{B2}$$

Since

$$\frac{h_0}{\mathfrak{a}\ell} = \frac{4\chi\tau}{(1-\chi)^2[(1-\chi)^2+\tau]} \le \kappa := \frac{4\chi}{(1-\chi)^2} = o(1), \tag{B3}$$

the roots satisfy $0 < \beta \le \min(\mathfrak{a}, \ell)$ and $\gamma \ge \max(\mathfrak{a}, \ell)$ for small ρ. The estimates also use

$$\beta \ge (1-\sqrt{\kappa})\min(\mathfrak{a}, \ell). \tag{B4}$$

To see this, set $t = (1-\sqrt{\kappa})\min(\mathfrak{a}, \ell)$. Both $\mathfrak{a} - t \ge \sqrt{\kappa}\,\mathfrak{a}$ and $\ell - t \ge \sqrt{\kappa}\,\ell$, so $(\mathfrak{a}-t)(\ell-t) \ge h_0$; the smaller root of this characteristic equation is β.

For $0 \le \tau \le 1/2$, the spectral separation satisfies

$$g := \mathfrak{a} - \ell = \frac{(1+\chi)^2-\tau}{2} \ge \frac{1}{4}. \tag{B5}$$

Define the nonnegative slow-mode weight

$$Q = \frac{(\gamma-\mathfrak{a})V + (\gamma-\ell)W + (r+c)R}{d}. \tag{B6}$$

The exact identity

$$(v, w)e^{AM}\begin{pmatrix} v \\ w \end{pmatrix} = Qe^{-\beta A} + (1-Q)e^{-\gamma A} \tag{B7}$$

and $\gamma - \mathfrak{a} = h_0/[(d+g)/2] \le h_0/g$ imply

$$\begin{aligned} Q &= W + \frac{(\gamma-\mathfrak{a})(V-W) + (r+c)R}{d} \\ &\le 4p + \frac{h_0}{g^2} + \frac{2\sqrt{p}(r+c)}{g} \le C_{\mathrm{b}}[p + \chi\tau + \sqrt{p\tau}]. \end{aligned} \tag{B8}$$

In particular $Q \le C_{\mathrm{b}}\sqrt{\rho} < 1$. Since $\gamma \ge \mathfrak{a}$,

$$\mathcal{E} \le [Qe^{-\beta A} - Pe^{-\mathfrak{a}A}]_+, \qquad P = V - W + Q = 1 + O(\sqrt{\rho}). \tag{B9}$$

Put $\theta = \beta/\mathfrak{a}$. Equations (B5) and (B2) give

$$\frac{1+\tau}{2} - C_{\mathrm{b}}\chi \le \theta \le \frac{1+\tau}{2}. \tag{B10}$$

For the upper inequality one can use $\ell/\mathfrak{a} = (1+\tau)/2 - \chi/(1+\chi^2) - \tau\chi^2/[2(1+\chi^2)]$. The lower inequality follows by subtracting $(\gamma-\mathfrak{a})/\mathfrak{a} = O(\chi\tau)$.

For $Q > 0$, the maximum of $Qx^\theta - Px$ over $x = e^{-\mathfrak{a}A} \in [0, 1]$ occurs in the interior because $\theta Q < P$. Its exact value is

$$D_{\max} = (1-\theta)\theta^{\theta/(1-\theta)}Q^{1/(1-\theta)}P^{-\theta/(1-\theta)}. \tag{B11}$$

For $Q = 0$, the envelope vanishes. Equation (B11) optimizes over all overall rates A, including unbounded-rate sequences.

2. Large rate ratios

For $\tau \ge 1/2$, small ρ ensures

$$\beta \ge 7/10, \qquad 1 \le \mathfrak{a} \le 101/100, \tag{B12}$$

by (B4). To obtain a bound uniform near repeated rates without dividing by d, introduce the positive convolution

$$J(s) = \int_0^s e^{-\beta(s-t)}e^{-\gamma t}\,dt \le se^{-\beta s}. \tag{B13}$$

When $d > 0$, also $J(s) \le d^{-1}e^{-\beta s}$. Matrix exponentiation or variation of constants gives

$$(e^{sM})_{12} = rJ(s), \qquad (e^{sM})_{21} = cJ(s), \tag{B14}$$

$$(e^{sM})_{11} - e^{-\mathfrak{a}s} = h_0\int_0^s e^{-\mathfrak{a}(s-t)}J(t)\,dt \ge 0, \tag{B15}$$

$$0 \le (e^{sM})_{22} \le e^{-\beta s}. \tag{B16}$$

The last inequality follows from the two nonnegative spectral weights, whose sum is one, and extends continuously to $d = 0$.

On $1/2 \le \tau \le 4$, the coefficients obey $h_0 \le C_{\mathrm{b}}\chi$ and $r+c \le C_{\mathrm{b}}$. Equations (B12)–(B15) give respective bounds $C_{\mathrm{b}}\chi s^2e^{-7s/10}$ and $C_{\mathrm{b}}se^{-7s/10}$ for the correction in (B15) and for $(r+c)J(s)$. On $\tau \ge 4$, small ρ gives $d \ge \ell - \mathfrak{a} \ge \tau/4$, while $h_0 \le C_{\mathrm{b}}\chi\tau$ and $r + c \le C_{\mathrm{b}}\sqrt{\tau}$. The corresponding bounds are $C_{\mathrm{b}}\chi se^{-7s/10}$ and $C_{\mathrm{b}}\tau^{-1/2}e^{-7s/10}$. Absorbing the polynomial factors therefore gives, uniformly on the entire large-ratio region,

$$\begin{aligned} (e^{sM})_{11} &\le e^{-\mathfrak{a}s} + C_{\mathrm{b}}\chi e^{-3s/5}, \\ (r+c)J(s) &\le C_{\mathrm{b}}e^{-3s/5}, \qquad (e^{sM})_{22} \le e^{-7s/10}. \end{aligned} \tag{B17}$$

Consequently,

$$\mathcal{E} \le [-Ve^{-\mathfrak{a}A} + C_{\mathrm{b}}\sqrt{\rho}\,e^{-3A/5}]_+. \tag{B18}$$

Optimize over $x = e^{-\mathfrak{a}A}$, with exponent $t = 3/(5\mathfrak{a})$. The maximum is at most $C_{\mathrm{b}}\rho^{1/[2(1-t)]}$. Since $\mathfrak{a} \le 101/100$, this exponent is at least $101/82 > 6/5$. Hence

$$\sup_{\tau \ge 1/2, A \ge 0} \mathcal{E} \le C_{\mathrm{b}}\rho^{6/5} = o\big(\rho/\log(1/\rho)\big). \tag{B19}$$

This estimate includes coincident decay rates and arbitrarily large B/A.

3. Intermediate and small rate ratios

Set

$$L = \log(1/\rho), \qquad \eta = L^{-1/2}. \tag{B20}$$

For $\eta \le \tau \le 1/2$, Eqs. (B8) and (B10) yield $Q \le C_{\mathrm{b}}\sqrt{\rho}$ and $1/(1-\theta) \ge 2/(1-\tau+C_{\mathrm{b}}\sqrt{\rho})$. All remaining factors in (B11) are uniformly bounded. Thus

$$D_{\max} \le C_{\mathrm{b}}(C_{\mathrm{b}}\sqrt{\rho})^{2/(1-\tau+C_{\mathrm{b}}\sqrt{\rho})} \le C_{\mathrm{b}}\rho e^{-\sqrt{L}/2}. \tag{B21}$$

For the last step, the logarithm after division by ρ is at most $C_{\rm b} - L(\tau - C_{\rm b}\sqrt{\rho})/(1-\tau+C_{\rm b}\sqrt{\rho}) \le C_{\rm b} - \sqrt{L}/2$.

On $0 \le \tau \le \rho^{1/4}$, including $B = 0$, the finer form of (B8) gives $Q \le C_{\rm b}\rho^{5/8}$. Since $1/(1-\theta) \ge 2 - C_{\rm b}\sqrt{\rho}$,

$$D_{\max} \le C_{\rm b}\rho^{5/4 - C_{\rm b}\sqrt{\rho}} \le C_{\rm b}\rho^{6/5}. \tag{B22}$$

This estimate also permits $\theta < 1/2$, which is possible for mixed stationary states.

4. Leading asymptotics

It remains to treat

$$\rho^{1/4} \le \tau \le L^{-1/2}. \tag{B23}$$

Here $\chi \le C_{\rm b}\sqrt{\rho} = o(\tau)$. The two coupling terms in (B8) admit the sharper estimates

$$\begin{aligned} \frac{h_0}{g^2} &= \frac{8\chi(1+\chi^2)\tau}{(1-\chi)^2[(1+\chi)^2-\tau]^2} \\ &\le 8\sqrt{\nu}\,\tau[1+O(\eta+\sqrt{\rho})], \\ \frac{2\sqrt{p}(r+c)}{g} &= \frac{4\sqrt{p\tau}(1+\chi)^2}{(1-\chi)[(1+\chi)^2-\tau]} \\ &\le 4\sqrt{p\tau}[1+O(\eta+\sqrt{\rho})]. \end{aligned} \tag{B24}$$

Therefore

$$Q \le 4p + [1+O(\eta+\sqrt{\rho})](4\sqrt{p\tau} + 8\sqrt{\nu}\,\tau) \le C_{\rm b}\sqrt{\rho\tau}. \tag{B25}$$

Squaring this bound while retaining the contribution from stationary-state tilt gives

$$\frac{Q^2}{4} \le 4p\tau + o(\rho\tau) \tag{B26}$$

uniformly. The remaining square and cross terms are bounded by constant multiples of $\rho\tau\sqrt{\eta}$, $\rho^{3/2}\sqrt{\tau}$, and ρ^2. After division by $\rho\tau$, these tend to zero uniformly by (B23); the relative error in the retained term is $O(\eta+\sqrt{\rho})$.

Write $m = 1/(1-\theta)$. Equation (B10) gives

$$m - 2 \ge 2\tau - C_{\rm b}\chi > 0, \qquad m = 2 + o(1). \tag{B27}$$

For $Q > 0$, use $Q \le Q_0 = C_{\rm b}\sqrt{\rho\tau} < 1$ to obtain

$$\begin{aligned} Q^{m-2} \le Q_0^{m-2} &\le \exp[(2\tau - C_{\rm b}\chi)\log Q_0] \\ &\le [1+o(1)]e^{-\tau L}. \end{aligned} \tag{B28}$$

Indeed, $\log Q_0 = -L/2 + (\log\tau)/2 + \log C_{\rm b}$ and $|\log\tau| \le L/4$, so the excess exponent above $-\tau L$ is at most $C_{\rm b}\eta + C_{\rm b}\sqrt{\rho}L = o(1)$. The prefactor in (B11) satisfies

$$(1-\theta)\theta^{\theta/(1-\theta)}P^{-\theta/(1-\theta)} = \tfrac{1}{4}[1+o(1)]. \tag{B29}$$

Equations (B26)–(B29) now imply

$$\mathcal{E} \le D_{\max} \le [4p + o(\rho)]\tau e^{-\tau L} \le \frac{4p/e + o(\rho)}{L}, \tag{B30}$$

using $\sup_{\tau\ge 0}\tau e^{-\tau L} = 1/(eL)$. Equations (B19), (B21), (B22), and (B30) together prove Proposition 4. Every remainder is uniform in the rates, their ratio, and the allocation of the stationary error budget. □

5. Matching construction at fixed error

The generator, kernels, and measured probabilities are given in (17)–(23). Their asymptotics establish the matching lower bound at fixed actual disturbance.

The parameter choice (17) gives

$$\begin{aligned} F = \frac{4p}{eL}, \qquad S &= \left(\frac{2}{e} + o(1)\right)\sqrt{\frac{p}{L}}, \\ \sqrt{AB}\,J_0 &= \frac{2/\sqrt{L}}{1-1/L}(S-F). \end{aligned} \tag{B31}$$

Hence the negative cross term in (22) is $(8/e + o(1))p/L$, while its positive population term is $(4/e + o(1))p/L$ and $w^2 S = o(p/L)$. Thus

$$\begin{aligned} q &= -\left(\frac{4}{e} + o(1)\right)\frac{p}{L}, \\ \epsilon(L) &= p[1+o(1)], \\ \log\frac{1}{\epsilon(L)} &= L + o(1). \end{aligned} \tag{B32}$$

Thus the construction attains the leading coefficient in Theorem 1 at its actual measured disturbance.

The construction realizes every sufficiently small positive disturbance. The function $\epsilon(L)$ is continuous, positive for all sufficiently large L, and tends to zero. Its range on a connected tail therefore contains an interval $(0, \epsilon_0]$. Any sequence of parameter values realizing disturbances that tend to zero must have $L \to \infty$, since $\epsilon(L)$ has a positive minimum on each compact subinterval of that tail. The lower asymptotic bound therefore holds throughout a punctured neighborhood of zero. Together with (35), this completes the proof of Theorem 1.

Appendix C: Diagonal stationary states

This appendix proves Theorem 2 using the full mixed-state equations (A7)–(A8). For a stationary state diagonal in the measurement basis, the population ν satisfies $\nu = a\nu + \epsilon(1-\nu)$, which gives (25). Only $q < 0$ contributes to the reversal, and then $0 < \nu \le \epsilon$. For small ϵ, the stationary state has $\nu < 1/2$, $|e\rangle = |0\rangle$, $|g\rangle = |1\rangle$, and $p = 0$. The input difference starts at $(u, z) = (1, 0)$ and gives $q = u(1)$. If $A = 0$ or $B = 0$, the Gram constraints decouple population from coherence, giving $q = e^{-\alpha} \ge 0$. It therefore suffices to consider $A, B > 0$.

1. Feedback bound

Let $(X, Z)^T = e^{tK}(1, 0)^T$. Lemma 6 gives $u(t) \ge 2e^{-\alpha t} - X(t)$. Introduce dimensionless time $x = \alpha t$ and

the parameters

$$\tau = \frac{B}{A}, \qquad \zeta = \frac{\lambda}{\alpha} = \frac{(1-\chi)^2+\tau}{2(1+\chi^2)}, \\ h_f = \frac{r_0 c_0}{\alpha^2} = \frac{2\chi\tau}{(1-\chi)^2(1+\chi^2)}. \tag{C1}$$

With $F(x) = e^{-x}$ and convolution on the positive half-line, the comparison population satisfies

$$X = F + h_f(F * e^{-\zeta\,\cdot} * X), \\ q \geq 2F(x) - X(x) \quad \text{at } x = \alpha. \tag{C2}$$

Varying A covers every $x \geq 0$ at fixed ν, τ.

For $\tau \geq 1/2$ and $\chi \leq 1/100$, put $k = 2/3$. Direct substitution gives

$$\eta_f := \frac{h_f}{(1-k)(\zeta-k)} \\ = \frac{36\chi\tau}{(1-\chi)^2(3\tau-1-6\chi-\chi^2)} \\ \leq 48\chi < 1. \tag{C3}$$

On any finite interval, the supremum norm of $e^{kx}X(x)$ in (C2) is bounded by $1/(1-\eta_f)$. Substituting this bound into the convolution gives

$$0 \leq X(x) - F(x) \leq \frac{\eta_f}{1-\eta_f} e^{-kx} \leq 100\chi e^{-2x/3}. \tag{C4}$$

Consequently, for sufficiently small χ,

$$(-q)_+ \leq \sup_{x\geq 0}[100\chi e^{-2x/3} - e^{-x}]_+ \\ \leq \frac{4}{27}(100\chi)^3 = O(\nu^{3/2}). \tag{C5}$$

This weighted-kernel argument is uniform at coincident decay rates and as $\tau \to \infty$.

For $0 < \tau \leq 1/2$, define

$$d_{\rm sp} = \sqrt{(1-\zeta)^2 + 4h_f}, \\ \beta = \frac{1+\zeta-d_{\rm sp}}{2}, \qquad \beta_+ = \frac{1+\zeta+d_{\rm sp}}{2}, \\ d_0 = \frac{d_{\rm sp}+\zeta-1}{2d_{\rm sp}}. \tag{C6}$$

For small ν, $0 < \beta < 1 \leq \beta_+$ and $0 < d_0 < 1$. The exact comparison solution is

$$X(x) = (1-d_0)e^{-\beta_+ x} + d_0 e^{-\beta x}.$$

Equation (C2) therefore gives $q \geq (1+d_0)e^{-x} - d_0 e^{-\beta x}$. Maximizing its negative part over $x \geq 0$ gives

$$(-q)_+ \leq \mathcal{D}_0(\beta, d_0) := (1-\beta)d_0 \left(\frac{d_0\beta}{1+d_0}\right)^{\beta/(1-\beta)}. \tag{C7}$$

Because $1 - \zeta \geq 1/4$ throughout this ratio range, the expansions

$$\beta = \frac{1+\tau}{2} + O(\chi), \qquad d_0 = \frac{8\chi\tau}{(1-\tau)^2}[1+O(\chi)] \tag{C8}$$

are uniform. They follow, for example, by rationalizing $d_{\rm sp} - (1-\zeta) = 4h_f/[d_{\rm sp} + (1-\zeta)]$.

2. Uniform asymptotics

Here $C_{\rm d}$ denotes a finite universal constant, independent of the rates and ν.

If $0 < \tau \leq \sqrt{\chi}$, then $d_0 \leq C_{\rm d}\chi^{3/2}$ and $\beta \geq 1/3$ for small χ. Hence

$$\mathcal{D}_0(\beta, d_0) \leq d_0^{1/(1-\beta)} \leq d_0^{3/2} = O(\chi^{9/4}) \\ = o\big(\nu/\log^2(1/\nu)\big). \tag{C9}$$

This bound controls the smallest rate ratios before the expansion involving $\log\tau$.

For the remaining range $\sqrt{\chi} \leq \tau \leq 1/2$, put $L = \log(1/\nu)$. Then $|\log\tau| = O(L)$ and $\chi L \to 0$. The sensitivity identity

$$\partial_\beta \log \mathcal{D}_0(\beta, d_0) = \frac{\log[d_0\beta/(1+d_0)]}{(1-\beta)^2} \tag{C10}$$

shows that the $O(\chi)$ correction to β in (C8) changes $\log\mathcal{D}_0$ by a uniformly $o(1)$ amount. The relative $O(\chi)$ correction to d_0 also has an $o(1)$ effect, since $\partial_{\log d_0}\log\mathcal{D}_0$ is bounded in this range. Using $\chi^2 = \nu/(1-\nu)$ gives

$$\frac{\mathcal{D}_0(\beta, d_0)}{\nu} = [1+o(1)]\mathcal{F}(\tau)\exp\left[-\frac{\tau L}{1-\tau}\right], \tag{C11}$$

where

$$\mathcal{F}(\tau) = \frac{1-\tau}{2}\left(\frac{1+\tau}{2}\right)^{(1+\tau)/(1-\tau)}\left(\frac{8\tau}{(1-\tau)^2}\right)^{2/(1-\tau)}. \tag{C12}$$

The function has a bounded continuous extension to $[0, 1/2]$, with

$$\mathcal{F}(\tau) = 16\tau^2\{1 + O[\tau(1+|\log\tau|)]\}. \tag{C13}$$

For $\tau \geq L^{-1/2}$, (C11) is exponentially small in $\sqrt{L}$. On the remaining range, the relative error in (C13) tends uniformly to zero. As $1/(1-\tau) \geq 1$,

$$\sup_\tau \frac{\mathcal{D}_0(\beta, d_0)}{\nu} \\ \leq [1+o(1)]\sup_{\tau>0} 16\tau^2 e^{-\tau L} + o(L^{-2}) \\ = \frac{64/e^2 + o(1)}{L^2}. \tag{C14}$$

Together with (C5) and (C9), this proves a uniform bound for all finite generators in the restricted class. For $q < 0$, Eq. (25) gives $\nu \leq \epsilon$. Since $x/\log^2(1/x)$ increases for small positive x, the same leading bound holds with ϵ in place of ν; for $q \geq 0$, it is immediate. The resulting bound depends only on the measured probabilities and extends to $\mathcal{Q}_{\rm diag}$ as in Section V.

3. Matching generator

Take $\nu \downarrow 0$, $\chi = \sqrt{\nu/(1-\nu)}$, $L = \log(1/\nu)$, and $\tau = 2/L$. Let $B = \tau A$, $U = \chi^2 A$, and in the measurement

basis choose

$$J = \begin{pmatrix} -\sqrt{B} & -\sqrt{U} \\ \sqrt{A} & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 0 & i\kappa_H \\ -i\kappa_H & 0 \end{pmatrix}, \qquad \kappa_H = \frac{\sqrt{AB}(\nu + \chi/2)}{1 - 2\nu}. \tag{C15}$$

Equation (A6) confirms that $\mathrm{diag}(\nu, 1-\nu)$ is stationary. The coefficients satisfy $\widetilde{C} = -c_0$, $R_0 = r_0$, $E = -\sqrt{AU}$, and $\omega = 0$. The subspace with real coherence is invariant and obeys (27). This single-jump generator realizes the negative-feedback mechanism described in Section IV.

Write $\zeta_+ = \lambda_+/\alpha$ and define

$$\Delta_- = \sqrt{(1-\zeta_+)^2 - 4h_f}, \qquad \gamma_\pm = \frac{1 + \zeta_+ \pm \Delta_-}{2}, \qquad d_- = \frac{1 - \zeta_+ - \Delta_-}{2\Delta_-}. \tag{C16}$$

For $\tau = 2/L$ and small ν, these quantities are real, with $0 < \gamma_- < \gamma_+$ and $d_- > 0$. The exact response is

$$q(x) = (1 + d_-)e^{-\gamma_+ x} - d_- e^{-\gamma_- x}, \qquad x = \alpha. \tag{C17}$$

Choose the finite positive scale

$$x_* = \frac{\log[(1+d_-)\gamma_+/(d_-\gamma_-)]}{\gamma_+ - \gamma_-}, \qquad A = \frac{x_*}{1 + \chi^2}. \tag{C18}$$

Then the achieved negative determinant is $-q(x_*) = \mathcal{D}_0(\beta_-, d_-)$ with $\beta_- = \gamma_-/\gamma_+$. These parameters have the same leading expansions as (C8). Substitution of $\tau = 2/L$ into (C11)–(C13) therefore gives

$$-q(x_*) = \left(\frac{64}{e^2} + o(1)\right)\frac{\nu}{L^2}. \tag{C19}$$

The actual disturbance is $\epsilon(\nu) = \nu[1 - q(x_*)] \sim \nu$. It is positive, continuous, and tends to zero, so the connected-range argument used for (23) covers every sufficiently small positive ϵ. This matches the upper bound (C14) and completes Theorem 2. Figure 2 shows both constructions approaching their asymptotic constants.

The curves in Figs. 1 and 2 evaluate the explicit solutions with 100-digit arithmetic. Figure 1 matches the two constructions at common actual errors by solving for their family parameters; Fig. 2 uses the actual error of each sampled realization for both axes.

Appendix D: Relation to the four-parameter family

This appendix establishes the relation stated in Section VI between the attaining constructions and the family used for numerical boundary searches in Ref. [1]. That family has

$$H_{\mathrm{old}} = X, \qquad J_{\mathrm{old}} = \sqrt{\gamma}\,|\psi_o\rangle\langle\psi_i|, \qquad |\psi_j\rangle = \begin{pmatrix} \cos\theta_j \\ i\sin\theta_j \end{pmatrix}, \qquad \gamma, t > 0. \tag{D1}$$

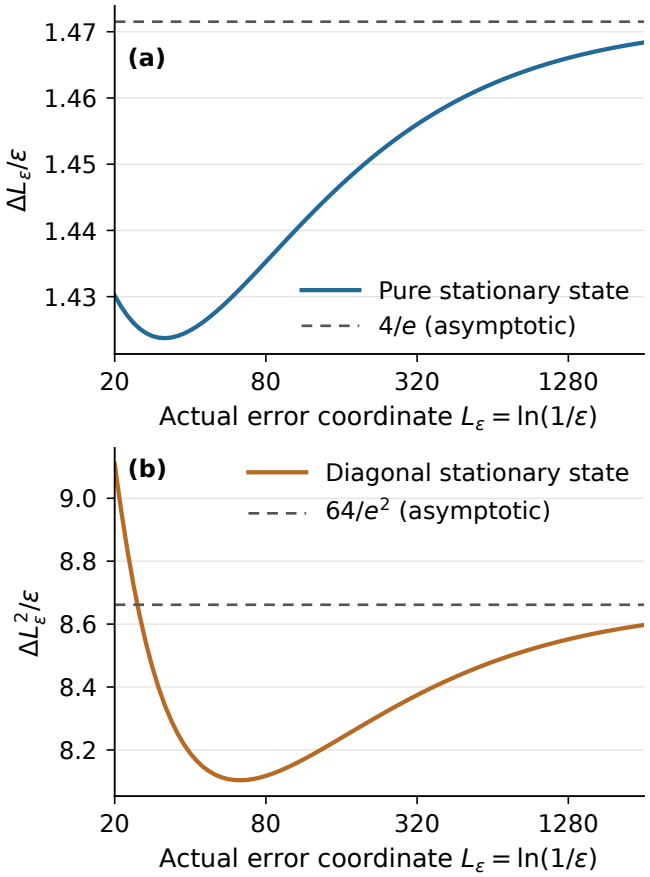


FIG. 2. Convergence of the explicit reset constructions, with population-order reversal $\Delta = \epsilon - a$ and actual disturbance $\epsilon = P(0|1)$. (a) The pure-stationary-state family, Eqs. (17)–(19), approaches $4/e$ after logarithmic rescaling. (b) The diagonal-stationary-state family, Eqs. (C15)–(C18), with $p = 0$, approaches $64/e^2$ after rescaling by $\log^2(1/\epsilon)$. Solid curves evaluate the explicit solutions; dashed lines show the asymptotic constants. The horizontal axes are logarithmic.

The rate and duration are unrestricted parameters of this family.

For each fixed finite parameter choice of the pure-stationary-state construction,

$$J = \sqrt{A+B}\,|u\rangle\langle e|, \qquad |u\rangle = \frac{-\sqrt{B}|e\rangle + \sqrt{A}|g\rangle}{\sqrt{A+B}}. \tag{D2}$$

The vectors u, e are real in the measurement basis. The unitary $S = \mathrm{diag}(1, i)$ preserves the preparation and measurement projectors and maps any normalized real vector to the form in (D1). Set $V_{\mathrm{old}} = S|u\rangle\langle e|S^\dagger$. For $t_n \downarrow 0$ and $\gamma_n t_n = A + B$, the channels in the prior family satisfy

$$\exp(-it_n[X, \cdot] + (A+B)\mathcal{D}[V_{\mathrm{old}}]) \longrightarrow \exp\big(\mathcal{D}[SJS^\dagger]\big). \tag{D3}$$

Continuity of the finite-dimensional matrix exponential places each such probability table in the closure of the prior family. This approximation holds at fixed A, B, p; the subsequent limit $L \to \infty$ approaches reset.

For the diagonal construction (C15), $\tau = 2/\log(1/\nu) > 4\chi$ for sufficiently small ν. The real jump then has a negative real eigenvalue ξ satisfying

$$\xi^2 + \sqrt{B}\,\xi + A\chi = 0, \qquad J' = J - \xi I, \\ H' = H + \frac{i\xi}{2}(J - J^\dagger). \tag{D4}$$

This identity shift preserves the GKSL generator. Since J' is nonzero, real, and singular, it has rank one. Moreover,

$$H'_{12} = ik', \\ k' = \kappa_H - \frac{\xi}{2}(\sqrt{A} + \sqrt{U}) > 0, \\ SH'S^\dagger = k'X. \tag{D5}$$

Write $J' = s|u'\rangle\langle v'|$ with $s > 0$ and real unit vectors. After the same phase change, choose duration $t = k'$ and jump rate $\gamma = s^2/k'$ in the prior family. The integrated generator then equals the transformed time-one generator of the diagonal construction. Thus the diagonal family belongs to the prior ansatz for every sufficiently small ν.

Appendix E: Comparison of exclusion regions

To compare with the earlier analytic outer bound, denote the reset-side function in Theorem 1 of Ref. [1] by

$$g_R(a) = (2 - a)(2a + a^{1/10}). \tag{E1}$$

That theorem excludes tables when $a \le 10^{-6}$, $\epsilon > g_R(a)$, and a second condition separating them from the swap endpoint holds. Near reset that second condition holds automatically. Since $g_R(a) = 2a^{1/10}[1 + o(1)]$ is locally increasing, its inversion gives the necessary lower bound

$$a \ge (\epsilon/2)^{10}[1 + o(1)]. \tag{E2}$$

Equation (E2) is the necessary condition obtained by inverting the earlier exclusion bound.

To compare the excluded regions directly, fix $C > 4/e$ and choose $\epsilon_C > 0$ so that Theorem 1 gives $a \ge \epsilon[1 - C/\log(1/\epsilon)]$ below this threshold. Every table in

$$\mathcal{W}_C = \big\{(a, 1-\epsilon) : 0 < \epsilon < \min(\epsilon_C, e^{-4C}, 10^{-6}), \\ \epsilon/2 < a < 3\epsilon/4\big\} \tag{E3}$$

is excluded by the present bound. None is excluded by the earlier theorem or its exchanged-coordinate version. Indeed, for $0 < a < 1$, $g_R(a) > 2a(2 - a) > 2a > \epsilon$, so the earlier reset condition fails. After exchanging the coordinates, the first coordinate is near one and fails the small-coordinate premise. The additional excluded region therefore has nonzero area.

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